Alignment of NIH Alzheimer's Research Funding with State-Level Disease Burden in the United States (2019–2023): A State-Level Panel Analysis

0. Abstract

Alzheimer's Disease (AD) is a progressive neurodegenerative disorder characterized by a gradual loss of cognitive ability, memory, and bodily function. Public health trends across the United States (U.S.) have shown a growing burden posed by this disease. However, it remains unknown whether federal research funding allocation corresponds to the geographic distribution of this burden, as prior studies have largely relied on cross-sectional national data without examining state-level dynamics over time. This study asks: does NIH Alzheimer's Disease research funding align with state-level disease burden, or is it primarily driven by existing institutional capacity? Data from all 50 U.S. states over five years (2019–2023) were drawn from CDC WONDER, NIH RePORTER, the U.S. Census Bureau, and NORC Dementia DataHub. Fixed-effects and Ordinary Least Squares regression models were estimated to assess whether changes in mortality and population structure were adequately supported by NIH funding allocation over time. NIH AD funding was weakly associated with measures of disease burden (mortality rate: $r = 0.345$, $\beta = -68{,}498.70$, $p = 0.852$). Instead, existing well-resourced research universities ($p < 0.001$, $\beta \approx +\$23.6M$ per institution) and prior funding in an area were strong and statistically significant predictors ($R^2 = 0.824$). These findings indicate that because NIH AD funding allocation is driven more by previous funding trends and institutional capacity than by state-level public health need, there are inequities in the representativeness of federally funded AD research. Future funding should target underserved areas to better address disparities in disease outcomes.

Keywords: Alzheimer's Disease, Disease Burden, Machine Learning, Mortality, NIH Funding, Public Health Policy, Regression Analysis

1. Introduction

Alzheimer's disease (AD) is the 6th-ranked cause of death in the United States, with over 114,000 deaths recorded in 2023[1]. This serious health challenge affects millions of people and places an enormous emotional and financial strain on families and caregivers. In addition, the annual number of new cases is projected to double by 2050[2].

The federal government invests primarily through the National Institutes of Health (NIH) in AD research, allocating $3.9 billion in research grants in 2025 alone[3]. Although funding-based research findings in one region can still affect outcomes in another region, the exact location where funding is sent creates a significant impact on that region through expanded clinical trials and strengthened healthcare infrastructure[4]. According to the NIH, "The funding for the new exploratory centers will enable investigators at those locations to build optimal infrastructure and to develop crucial statewide partnerships… centers also will have unique scientific focuses based on their local interests and communities"[5].

Given the unique local focus that regional funding provides, it is important to consider the following question:

Does the amount of research funding reflect where the disease burden is the greatest?

This study hypothesizes that NIH research funding does not adequately reflect disease burden across the United States. This issue is increasingly relevant for AD, where significant disparities exist in both disease burden and access to research participation: Vulnerable populations in certain racial, socioeconomic, and geographic groups experience disproportionately higher risk but remain underrepresented in research and clinical trials, highlighting broader inequalities in how research systems operate[6]. Despite growing attention to these disparities[7], existing research has largely focused on cross-sectional or national-level analyses[8,9], often overlooking how funding and disease burden evolve over time or vary across smaller regions such as states. Because of the growing disease and death rates associated with AD[2,10], a more in-depth analysis of the effectiveness of current funding is needed.

Official U.S. death statistics show that age-adjusted AD mortality rates vary substantially by state: the following map from the CDC's National Vital Statistics System shows state-level differences in AD deaths per 100,000 population:

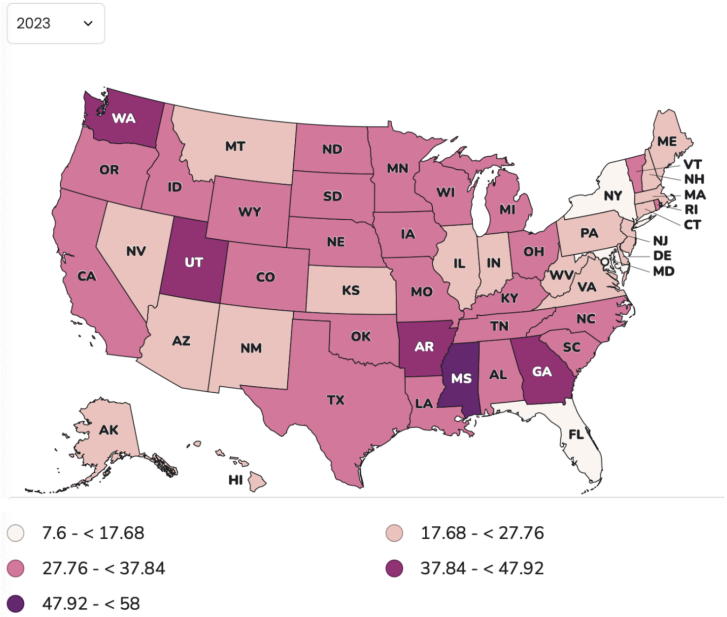

Figure 1. Geographic distribution of state-level Alzheimer's disease mortality rates in 2023, measured as age-adjusted deaths per 100,000 population. Higher mortality rates per state are shown with a darker shaded region, highlighting the differences in mortality across different regions. Data sourced from the Centers for Disease Control and Prevention (CDC).

As shown in Figure 1, the Southern region carries a disproportionately high AD mortality burden, with states such as Mississippi, Alabama, and Georgia recording age-adjusted death rates exceeding 37.84 per 100,000 population. In contrast, states like New York and Maryland, which cluster in the lowest mortality brackets (7.6–17.68 per 100,000), receive substantially greater NIH AD research investment due to their dense concentration of R1 research universities and established medical centers[9]. This disparity is reflected in regional funding figures: the Northeast receives approximately $97,804 in NIH AD funding per Alzheimer's-related death annually, while the South receives only $75,466 per death despite its considerably higher mortality burden (Figure 7). This forms a misalignment between where AD research is conducted and where the disease exerts its greatest toll, driven by institutional capacity rather than geographic health need.

The significance of this study lies in its potential to inform how federal research funding can be more effectively and equitably distributed. By evaluating whether funding patterns align with geographic need, this analysis provides insight into the efficiency of current allocation systems. The findings have practical implications for policymakers by identifying whether adjustments are needed to better target underserved regions: specifically, by producing state-level rankings of funding relative to disease burden that can directly guide NIH priority policies, grant review criteria, and the geographic targeting of new research center investments. Additionally, this study contributes to the field of public health by applying a longitudinal, state-level approach, offering a more precise understanding of funding dynamics over time. By noticing where burden and investment are not aligned, this regression framework can isolate which structural factors, such as the absence of R1 institutions or lower median incomes, are the main barriers to funding access, giving policymakers a concrete basis for targeted interventions such as infrastructure

grants or partnerships with regional medical schools. Ultimately, the results can support the development of more balanced funding strategies by providing evidence for need-based response formulas factor in both disease burden and institutional capacity, therefore improving access to research opportunities and promoting a more equitable public health impact.

This study focuses on a balanced panel of all 50 U.S. states over the period 2019–2023, using publicly available data on AD mortality, prevalence, population demographics, and NIH funding. The analysis is limited to state-level data and does not capture within-state variation, such as differences across counties or communities.

This study is grounded in a need-based framework[11], which suggests that funding should respond to indicators of disease burden, such as mortality or prevalence, to maximize public health impact. In contrast, a capacity-based framework would provide that funding flows toward regions with established research infrastructure, institutional expertise, and prior funding success.

Methodology Overview

This study uses a quantitative and longitudinal research design using a state-level panel dataset spanning from 2019-2023. Data were compiled from multiple public sources, including CDC mortality records, NIH funding databases, demographic data, and AD prevalence estimates. First, a five-year state-level panel dataset (2019–2023) was compiled by combining AD mortality, prevalence, population, and NIH funding data from multiple public sources. Then, after comparing the variables associated with funding and disease burden using correlation analysis, fixed-effects regression models were created using machine learning to examine whether changes in disease burden were associated with changes in research funding across states. Finally, these patterns were compared to assess whether NIH AD funding aligns more closely with public health needs or with existing research capacity.

2. Methodology

Data Sources
Data from four publicly available datasets were combined into a five-year panel covering all 50 U.S. states, as shown in Table 1.

Source	Data Type	Key Variables	Years Used
CDC WONDER[12]	Mortality (ICD-10 G30)	AD deaths, age-adjusted death rates	2019-2023
NIH RePORTER[13]	Research funding	Award amounts for projects tagged "Alzheimer's Disease" by awardee state	2019-2023
NORC Dementia DataHub[14]	Prevalence	Estimated number of adults (65+) living with AD	2019-2023
U.S. Census Bureau[15]	Population	State population estimates for normalization	2019-2023

Table 1. Dataset sources of AD Disease burden and NIH funding

All dollar amounts were converted to 2024 U.S. dollars using the Consumer Price Index (CPI) from the Bureau of Labor Statistics[16] to remove the effect of inflation.

Data Preparation and Key Definitions
Each dataset was cleaned and merged by state and year to form a single table (50 states × 5 years) using Python software. Any observation with a missing value was removed from the table.

Two main variables were created, 'disease burden' and 'research investment,' as described in Table 2.

Metric	What It Means	Formula	Why It Matters
Disease burden	Age-adjusted AD mortality rates and total annual deaths.	$TotalDeaths_{i,\square} = \Sigma Deaths_{i,\square}$ i = state, t = year	Mortality-based measures capture severe and clinically consequential AD disease outcomes and are consistently recorded across states, allowing reliable comparison of disease burden over time.
Research investment	The total NIH AD funding and funding	$MortalityRate_{i,\square} = (TotalDeaths_{i,\square} /$	Normalizing NIH funding by population or disease

	normalized per older adult (65+).	$TotalPopulation_{i,\square}$) * 100,000	outcomes helps distinguish differences in research investment intensity from differences driven solely by state size.

Table 2. Description of variables used in analysis

Statistical Analysis

As an initial step, Pearson correlations were calculated between state-level measures of AD disease burden and NIH AD research funding. This indicates whether states with higher disease burden receive higher levels of research investment, without accounting for confounding factors. Then, a correlation matrix was created to outline the strength of relationships between all of the variables studied.

Following the correlation matrix analysis, scatterplots were constructed to visually examine the relationship between NIH AD research funding and three key burden-related variables: age-adjusted mortality rate per 100,000 population, the size of the 65-plus population, and total annual deaths. In each scatterplot, total NIH AD funding (in millions of dollars) was plotted on the y-axis, with the respective burden variable on the x-axis, producing one data point per state per year across the 2019–2023 panel. Pearson correlation coefficients (r) and associated p-values were calculated for each pairing and displayed within the figures to indicate the strength and statistical significance of each relationship. To identify which specific states exhibited notable trends, such as receiving disproportionately high funding relative to their disease burden, or high burden relative to their funding, states were ranked by both their average annual NIH funding and their average mortality rate across the five-year period. States falling in the highest funding quartile but lowest mortality quartile (such as California, Massachusetts, and New York) and those in the highest mortality quartile but lowest funding quartile (such as Alabama, Mississippi, and Arkansas) were identified through this ranking process.

To examine broader national trends over time, three additional visualizations were produced and compiled in Figure 4. First, total NIH AD funding was aggregated across all 50 states for each year from 2019 to 2023 and plotted as a time series to capture the national funding trajectory. Second, the average age-adjusted AD mortality rate across all states was similarly plotted by year to show how national mortality trends evolved over the same period. Third, average NIH AD funding was calculated separately for each of the four U.S. census regions (Northeast, South, Midwest, and West) and displayed as a bar chart to illustrate regional disparities in research investment. These visualizations were produced using Matplotlib in Python.

The primary statistical approach was to design a **two-way fixed effects panel regression model**. This model controls for unobserved characteristics that are constant within each unit over time (e.g., states or counties) as well as shocks common to all units in a given year. This approach isolates the association between changes in NIH funding and changes in mortality within the same unit over time. The formula for this model is as follows:

$$\mathbf{Y_{i,\square} = \beta_1 Burden_{i,t} + \gamma X_{i,t} + \alpha_i \delta_t \varepsilon_{i,t}}$$

$\mathbf{Y_{i,}}$	Denotes NIH AD research funding in state i and year t
$\mathbf{Burden_{i,t}}$	Represents AD disease burden, measured using mortality-based indicators
$\mathbf{X_{i,t}}$	Vector of time-varying control variables
$\boldsymbol{\alpha_i}$	Captures state fixed effects, accounting for time-invariant differences between states (such as long-standing research infrastructure)
$\boldsymbol{\delta_t}$	Captures year fixed effects, controlling for national trends and shocks common to all states (such as changes in overall NIH budgets or federal policy)
$\boldsymbol{\varepsilon_{i,t}}$	The error term, which accounts for sudden changes in a state's research leadership, grant-writing capacity, or large one-off grant awards or rejections

Table 3. Explanation of formula components

This method helps ensure the results are not driven by factors unrelated to AD.

Software and Tools
I used **Python** for data cleaning and analysis:

- Pandas for merging and processing data
- Statsmodels for correlation and regression
- Matplotlib and Seaborn for creating maps and graphs

Ethical Considerations
All data used in this project come from publicly available, anonymized sources. No human subjects were involved, so Institutional Review Board (IRB) approval was not required. The project follows open-data standards, and all code has been made public on GitHub.

3. Results

As seen in Figure 2, the correlation matrix highlights how annual funding has a strong correlation with a number of variables. Using guidance that a 'strong' correlation is any value above 0.7[17], two variables stood out as particularly notable. First, annual funding is seen to have a strong correlation with the number of high research (R1) universities in that area, with a correlation value of 0.887. Even stronger is the correlation between the number of annual deaths and the size of the 65-plus population, with a 0.981 correlation value. These relationships indicate both research funding and disease burden are closely tied to underlying structural and demographic factors. In particular, the strong association between funding and R1 institutions suggests that the distribution of NIH funding is heavily influenced by existing research infrastructure, while the tight link between deaths and the older population reflects the expected demographic concentration of AD. Realizing these patterns allowed the creation of regression models to focus on R1 universities and age sectors as primary variables. However, given prior literature suggesting that wealthier states may have structural advantages in securing federal grants, median household income was also included as a secondary economic control variable to assess whether state-level wealth independently predicts research investment.

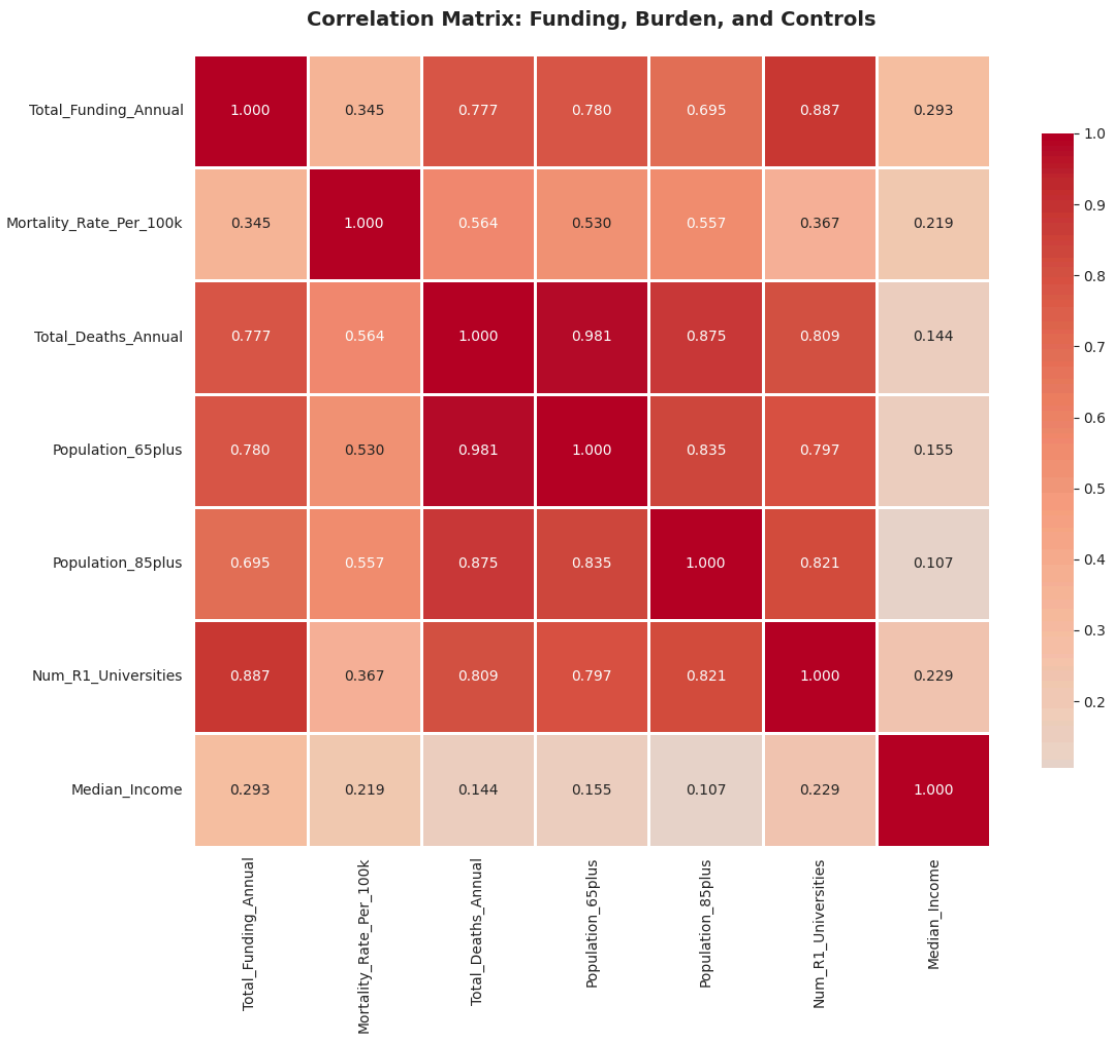

Figure 2. Correlation matrix measuring funding, burden, and controls. Higher Pearson correlation values and darker shading indicate stronger relationships between variables.

Statistical Findings

As shown in Figure 3, funding and mortality trends vary based on which variable is being studied. The maps and scatterplots show a clear pattern:

- California, Massachusetts, and New York receive much higher investment shares than their disease burden would predict, likely due to established research universities and medical centers.
- Alabama, Mississippi, and Arkansas have high AD mortality but low per-capita research funding, possibly indicating the low income per capita and the old age of these states.
- The geographic imbalance remains fairly consistent over the five years studied.

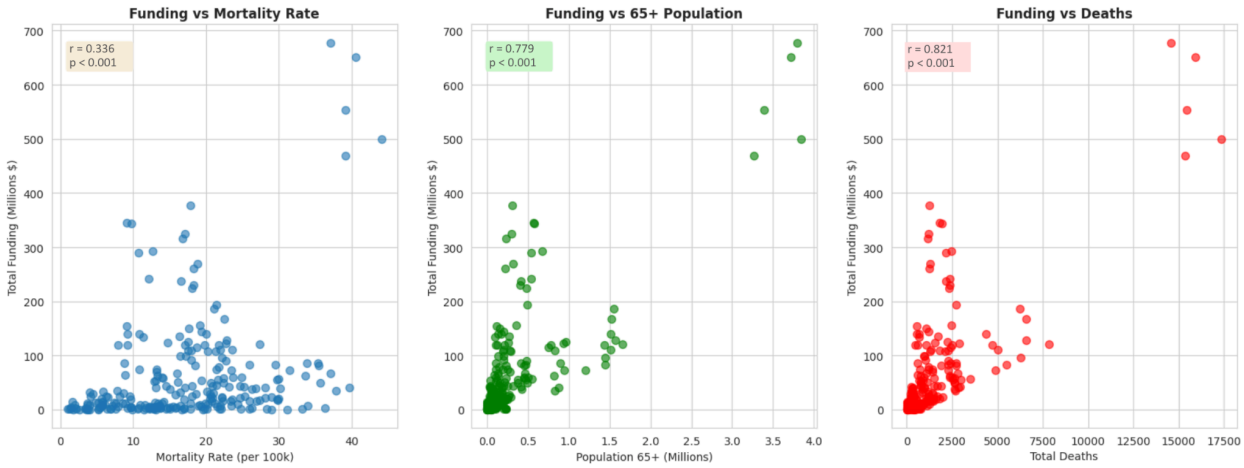

Figure 3. Statplot graphs of total funding vs. mortality rate, 65-plus population, and deaths, respectively.

As shown in Figure 4, national NIH AD funding increased steadily from 2019 to 2023, while national mortality rates fluctuated over the same period, further suggesting that funding growth was not directly responsive to changes in disease burden. Regional disparities in average funding also remained consistent across the five years studied.

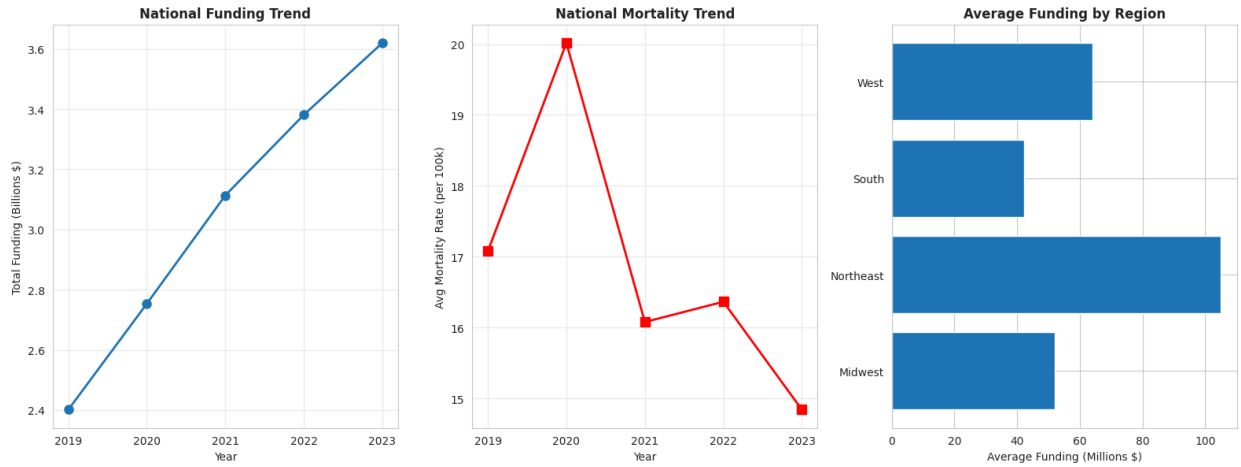

Figure 4. Broader trends between national funding, national mortality, and average funding by region.

Measure	Estimated Value	Interpretation
Adjusted R^2	≈ 0.8239	Funding variation is primarily explained by state and year fixed effects and prior funding, rather than contemporaneous disease burdens.
Association with Burden (Correlations)	Mortality Rate: r = 0.345 Population 65+: r = 0.780 Total Deaths: r = 0.777	Areas with higher populations of people 65+ and higher total deaths have a strong association with a higher burden. The mortality rate has a less pronounced relationship with the burden compared to the other factors. The "Population 65+" and "Total Deaths" variables are strong predictors of the burden.
Mortality Rate Coefficient	-68,498.70 (p = 0.8517)	As it is a statistically insignificant and negative coefficient, this relationship is substantially weaker than that observed for the population aged 65 and older (r=0.780) and total deaths (r=0.777), both of which strongly indicate overall disease burden.
Gini Coefficient	≈ 0.6830 ($p < 0.05$)	There is a high inequality in funding distribution. Changes in mortality-based disease burden within a state are not robust predictors of changes in NIH funding after controls.

Table 4. Interpretation of statistical measures

OLS Model Findings: Determinants of NIH Funding

A regression model was created employing heteroscedasticity-robust (HC3) standard errors[18] to ensure reliable inference. The results reveal a clear hierarchy among the determinants of state-level NIH AD research funding, with institutional research capacity emerging as the dominant driver.

The most influential predictor is the number of R1 universities within a state. This variable exhibits a large and highly statistically significant effect ($p<0.001$), with each additional R1 institution associated with an average increase of approximately $23.6 million in NIH funding. This finding strongly confirms that federally funded biomedical research is fundamentally capacity-led, with institutional research infrastructure as the primary conduit for federal investment.

Demographic effects display a notable divergence across age cohorts. The 'population aged 65 and older' variable has a positive and statistically significant association with NIH funding, with an estimated increase of $64.75 per additional individual in this age group ($p=0.007$). This suggests that broader demographic scale contributes to baseline funding allocations. In contrast, the population aged 85 and older exhibits a negative effect which is statistically significant. Each additional individual in this cohort is associated with a reduction of $621.49 in funding ($p=0.007$). These contrasting results imply that states with stronger concentrations of the "oldest old", who are more likely to require intensive clinical care rather than participate in research, do not receive commensurate increases in research funding.

Beyond institutional and demographic factors, the model also examined whether state-level economic conditions independently predict NIH funding allocation. Economic factors play a secondary but meaningful role. Median household income is a statistically significant positive predictor of NIH funding ($p=0.006$), with each $1 increase in median income associated with an approximate $485 increase in funding. While modest in magnitude, this effect suggests a stilt toward wealthier states, likely reflecting advantages in local health infrastructure, grant-writing capacity, and institutional support mechanisms.
Notably, measures of clinical disease burden do not influence funding levels. The state-level AD mortality rate is statistically insignificant ($p=0.350$, $p=0.350$), indicating that NIH funding is effectively decoupled from the intensity of AD-related deaths. This finding underscores the limited role of direct health need in shaping research investment patterns.

Figure 5 shows a weak alignment between AD mortality burden and NIH funding across states. States with higher mortality rates do not systematically receive higher funding per death, while several low-mortality states receive disproportionately large funding amounts. The wide dispersion of data points around the regression line indicates substantial variability, suggesting that increases in mortality burden are not reliably associated with increases in funding.

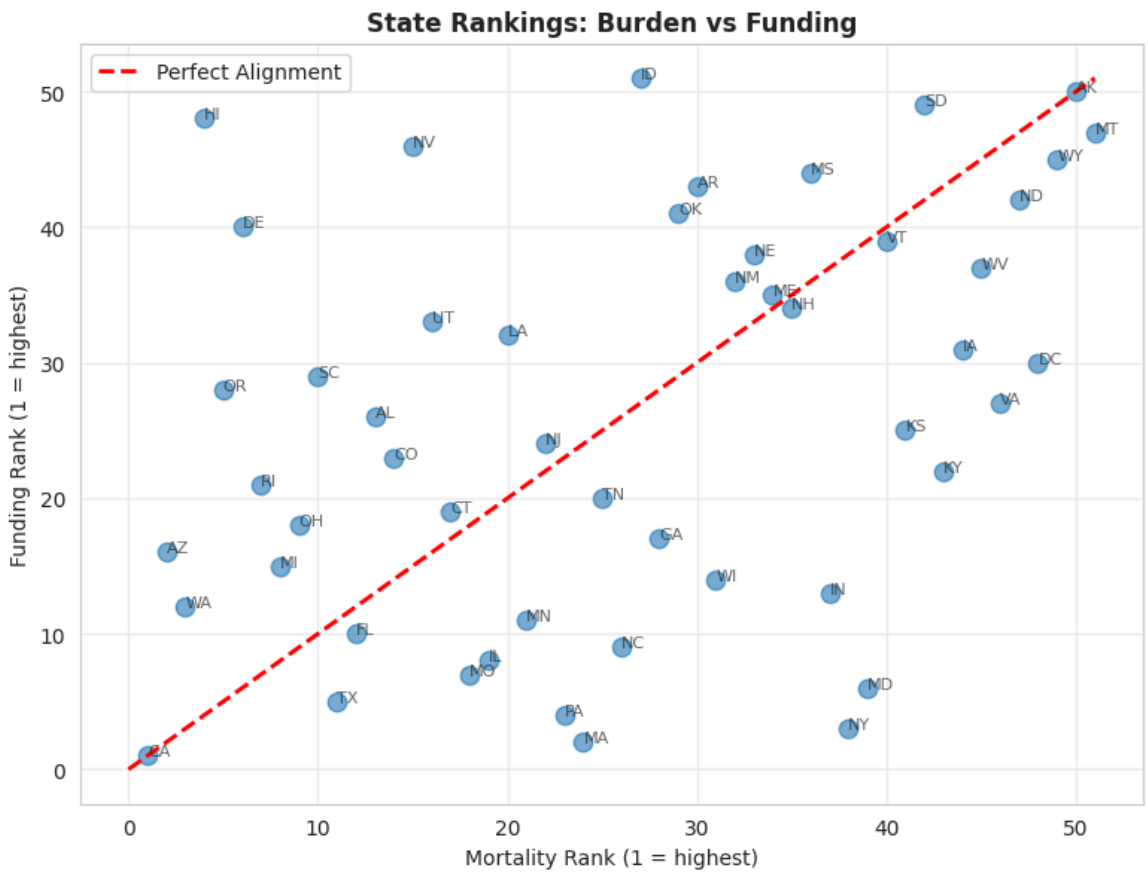

Figure 5. Statplot and logarithmic regression of disease burden vs. annual funding.

In Figure 6, the distribution of funding per death is highly right-skewed, with a median of approximately $38,623, indicating substantial dispersion. This suggests that while most states receive relatively moderate levels of funding per AD-related death, a small number of states receive disproportionately high funding amounts. This imbalance highlights significant inequality in the allocation of research resources, where funding is concentrated in select states rather than evenly distributed according to disease burden.

The large skew further supports a **capacity-driven** funding pattern, as states with established research infrastructure are more likely to attract large funding allocations, regardless of their relative disease burden.

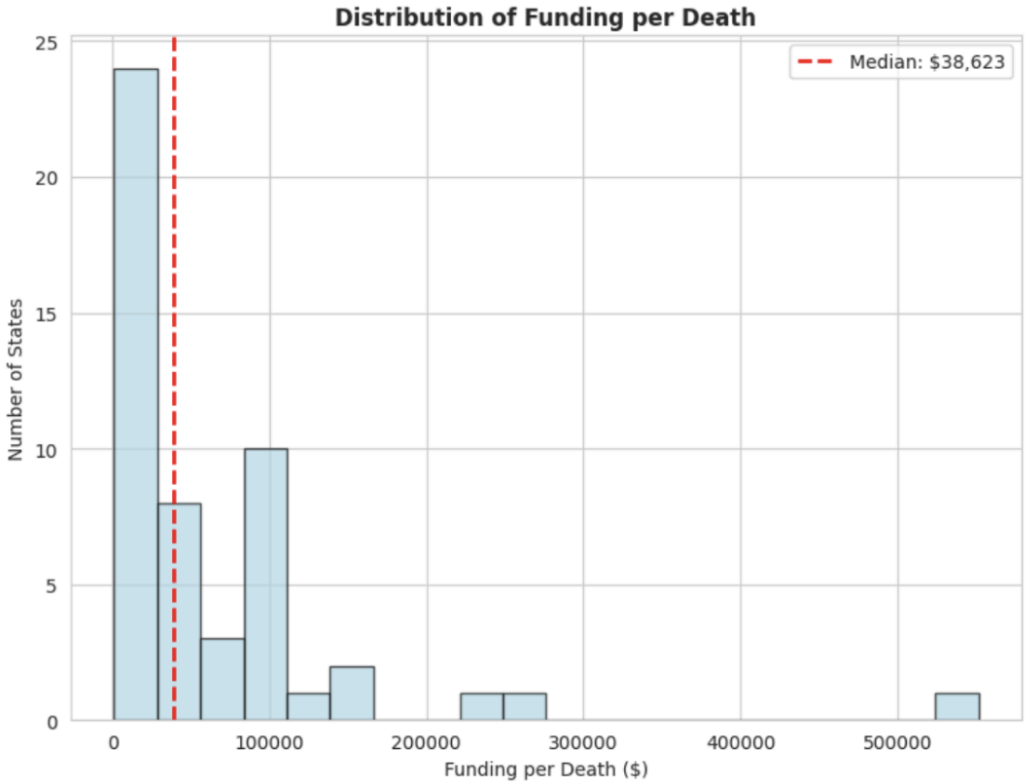

Figure 6. Bar graph distribution showing a positive skew between funding per death and state distribution.

Further, across both measures in Figure 7, the Northeast consistently receives the highest level of funding, with approximately $97,804 per death and $646 per elderly resident. The South and Midwest occupy intermediate positions, while the West receives substantially lower funding by both metrics, with only $26,737 per death and $209 per elderly resident. Notably, regions such as the South (where AD burden is relatively higher) do not receive proportionally higher funding, while the Northeast receives disproportionately greater resources relative to both deaths and elderly population size.

These regional differences in funding are not solely driven by demographic variation, but are likely influenced by underlying differences in research infrastructure and institutional capacity.

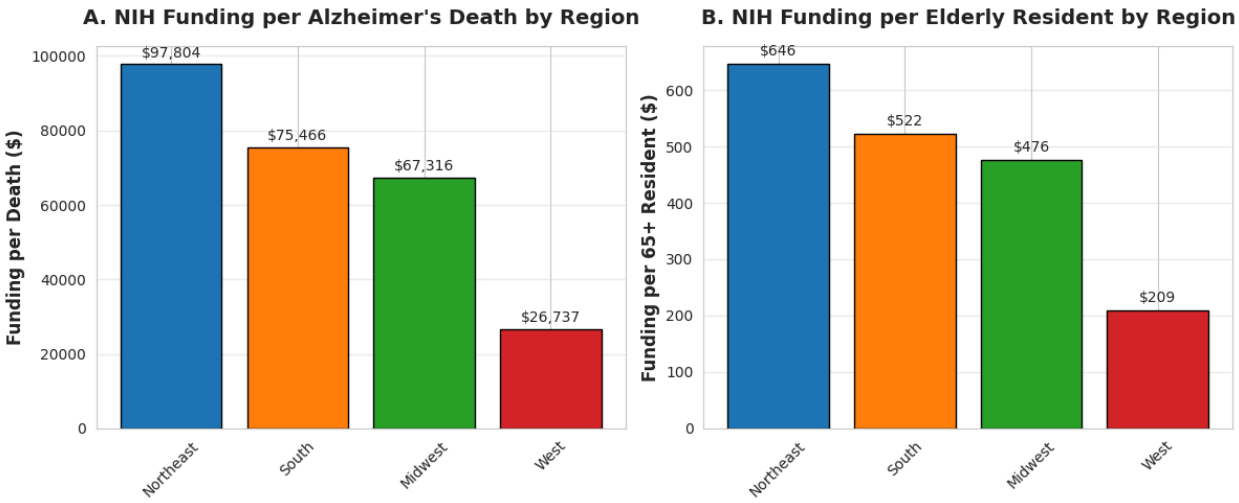

Figure 7. Regional distribution of Funding per Alzheimer's deaths, annually.

4. Discussion

This study asked whether NIH AD research funding aligns with state-level disease burden or is primarily driven by existing institutional capacity. The findings support the hypothesis that funding does not adequately reflect geographic disease burden. The analysis suggests that AD research funding does not perfectly align with where the disease burden is highest. Instead, it tends to follow where strong research infrastructure already exists.

Possible Explanations

1. **Concentration of Research Institutions:** NIH funding often goes to established research centers. States with fewer universities or medical schools are less likely to receive large grants.
2. **Historical Influence:** Funding builds over time; once a research program is established in a state, it tends to keep receiving grants, even if local disease rates change.
3. **Policy and Economic Factors:** Wealthier states may have more capacity to apply for and manage complex grants.

This imbalance means that the areas hardest hit by AD, which are often rural or lower-income regions, may have fewer opportunities for research participation, early diagnosis, and access to experimental treatments. Strengthening local research capacity in those areas could make national research efforts more representative and equitable. To better understand which of these explanations most fully accounts for these findings, future research should continue to investigate inequities in NIH-funded AD research across the United States.

Key Policy Implication

The results provide strong evidence of a capacity-led rather than needs-led funding regime. NIH AD research funding disproportionately favors states with dense research infrastructure and higher incomes, while failing to respond to regions with the greatest clinical vulnerability. The negative coefficient for the 85+ population and the statistical insignificance of mortality rates highlight a structural misalignment between federal research investment and the geographic distribution of mortality, raising important questions about equity and efficiency in national research funding policy.

Limitations of the Study

This study is subject to several limitations that should be considered. First, there is potential location bias, as NIH RePORTER assigns funding to the state of the primary research institution rather than the location of study participants, which may misrepresent where the benefits of research are realized. Second, AD reporting and diagnosis could vary across states, affecting the accuracy of prevalence and mortality data. Third, the use of state-level data presents the risk of ecological fallacy, as state-level relationships may not reflect individual-level dynamics between disease burden and research funding allocation. Fourth, the study may not incorporate all potentially relevant variables such as factors within healthcare access, education, or racial disparities due to data constraints. Fifth, using NIH grant data specifically could not fully show the way that funding is being used in other areas. Sixth, because this analysis is strictly correlational, it cannot determine conclusive causal relationships, potentially missing confounding variables like changes in NIH research policy. Seventh, although this study

accounts for funding lags, the temporal relationship between funding and outcomes may involve lags that vary per region and are not fully captured. Eighth, the relatively short study period (2019–2023) could limit detection of long-term relationships between funding and disease burden, given the gradual nature of both AD progression and research funding cycles. Finally, this analysis does not account for additional factors such as socioeconomic conditions, demographic differences, or healthcare access, which may influence both disease burden and the distribution of research funding but were excluded due to data and scope constraints.

Conclusion

This project provides evidence that AD research funding in the United States is only weakly aligned with actual disease burden across states, providing novel insights into the reality of regional differences in mortality and funding. While some correlation exists, much of the funding appears concentrated around existing research hubs rather than in regions with the highest need.

Improving the alignment between disease burden and funding allocation could enhance both the effectiveness and equity of national AD research efforts. Previous research indicates that location-based investments can improve health outcomes in a respective area, proving the potential benefits of a more need-responsive rather than capacity-based funding model. Overall, this study highlights the importance of policies aimed at strengthening research capacity in underserved regions to ensure that advances in AD research are more broadly distributed and accessible to all.

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6. Appendices

Appendix A – Code Repository
Full code and data processing steps will be available at:
GitHub Repository: [redacted]

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7. Author's Note

This research was completed as an independent project using publicly available data and open-source tools. It represents my personal effort to apply data science techniques to a real-world public health question and to explore how data can reveal inequalities that might otherwise go unnoticed.